SRI HARI N U

Software Engineer II – Backend, Distributed Systems & Applied AI

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PROFESSIONAL SUMMARY

Software Engineer with **4+ years of experience** at **JPMorgan Chase**, building high-throughput, scalable distributed systems for **Equity Derivatives Trade Management Platform**, with domain knowledge of regulatory assessment, trade lifecycle, corporate actions, cashflows and derivatives across Equity, FX, Credit and Commodities.

Specialized in Event Driven Architecture, Microservices, Distributed Systems, Performance Optimization, REST APIs, Data Pipelines, System-Design (LLD/HLD), CI/CD and Agentic AI workflows.

TECHNICAL SKILLS

Languages & Frameworks: Java, Spring Boot, Python, SQL, JavaScript, React
Data & Streaming: Kafka, Spark (PySpark), Avro, Parquet, Pandas
Database & Caching: PostgreSQL, MySQL, Sybase, pgvector, GemFire (Apache Geode)
Cloud & Infrastructure: AWS - EC2, ECS, S3, SQS, SNS, Lambda, Glue, Athena, Step Functions, Terraform
DevOps: Docker, Kubernetes, Jenkins, Spinnaker, Git, Datadog
AI Tools & Testing: LLMs, Claude Code, LangChain, RAG Pipeline, MCP, JUnit, Mockito, Pytest, Playwright

EXPERIENCE

JPMorgan Chase & Co. — *Bengaluru, India*

Software Engineer II — *Feb 2024 – Present*

- **Built and own core services of equity derivatives trade management,** processing **1M+ trade events/day** (Listed and OTC) across **a hybrid-cloud architecture** distributed across **North America, EMEA, APAC** regions, feeding downstreams - clearing, settlement, accounting, regulatory reporting, etc. through **data lake, Kafka, AMPS messaging, IBM MQ, and centralized file transfers.**
- Engineered **a real-time trade and position data ingestion pipeline** into **the firm-wide data lakehouse** using AWS Glue, PySpark, ECS, SQS, and Athena, writing partitioned **Parquet** data into **Apache Iceberg** tables on S3; implemented automated retries and end-of-day source audit reconciliation to ensure **data consistency and idempotency.**
- **Redesigned end-of-day regulatory report processing** using data segregation, batching, parallel processing, concurrent task execution, and API optimization, increasing throughput and generation time **by 86%** (3 hrs → 25 min), eliminating compliance SLA breaches.
- **Reduced transaction-model (CDM) creation time by 5x (10 min → 2 min)** for composite trades containing 5K+ positions by optimising data transformers, mapping logic, and model construction, **improving** throughput and preventing cascading queue during peak traffic.
- Delivered regulatory assessment for **six regimes** rewrite program - SEC, FCA, CFTC, ASIC, HKMA and JFSA, by implementing regulatory rules, reference data API integration, enriching existing trades and positions over **$5T notional scale**, achieving approximately **97% report acceptance** by DTCC.
- Implemented **Common Domain Model** (CDM)-based Digital Regulatory Reporting (DRR) for equity derivatives, contributing to the **firm's** FINOS Adoption Achiever Award**.**
- Developed a **RAG assistant** using React, FastAPI, an internal ADK built on LangChain, secured with OAuth 2.0 and JWT access-token validation. Grounded responses in Jira, Confluence and internal documentation to help Operations and Engineering teams **navigate over 45+ workflows**.
- Built **multiple copilot agents** to orchestrate **15+ microservices components** in local development environments, improving development speed by **20%.**

Consultant (via mThree) — *Sep 2022 – Jan 2024*

- Built **a high-throughput trade-feed consumer** to onboard a new desk feed clearing - implementing custom-optimized Avro deserializer, transformers and schema mappings, **achieving a 99% clearing rate**.
- **Modernized** 5 Java services (Java 11 → 21, Spring Boot 2.7 → 3.2) of complex codebase and in-house Terraform modules (infra), **strengthening applications and cloud infrastructure security.**
- Built a **custom Kafka Connect source connector** for insert-only CDC from Sybase event tables, using timestamp-based SQL polling to publish new events to Kafka for downstream consumers.
- Automated **end-of-day reconciliation** using scripts and Control-M (scheduler); ingested files into Sybase tables and enriched records through multi-table SQL joins, **eliminating manual position-break checks**.
- Built **an automated end-to-end regression suite** (using Pytest, AWS SDKs, Playwright) reducing manual testing/regression effort, **saving 8+ developer-hours weekly.**

EDUCATION

B.E. in Computer Science & Engineering, Visvesvaraya Technological University, Belagavi — *2018 – 2022*

CERTIFICATIONS & AWARDS

AWS Certified Cloud Practitioner (2024) | FINOS Adoption Achiever Award (2024)